

Experiment / Case Study No.: 9

1. Experiment Title:

Implement localStorage and sessionStorage to Save User Preferences Like Theme Settings

2. Software/Tools Required:

Visual Studio Code, HTML5, CSS3, JavaScript (Web Storage API — localStorage, sessionStorage), Google Chrome Browser / Microsoft Edge

3. Experiment Program Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Theme Preference</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <div class="header">
      <h1>🌟 Theme Preference</h1>
      <p class="subtitle">Choose your vibe and make it stick</p>
    </div>

    <div class="button-group">
      <button class="btn btn-light" onclick="setTheme('light')">
        <span class="icon">☀️</span>
        <span class="label">Light Theme</span>
      </button>
      <button class="btn btn-dark" onclick="setTheme('dark')">
        <span class="icon">🌙</span>
        <span class="label">Dark Theme</span>
      </button>
      <button class="btn btn-reset" onclick="clearTheme()">
        <span class="icon">🔄</span>
        <span class="label">Clear Preference</span>
      </button>
    </div>

    <div class="status-card">
      <p id="message">No theme selected yet</p>
    </div>

    <div class="storage-info">
      <div class="storage-item">
        <span class="storage-label">localStorage</span>
        <span class="storage-value" id="localValue">—</span>
      </div>
    </div>
  </div>
```

```

        <div class="storage-item">
            <span class="storage-label">sessionStorage</span>
            <span class="storage-value" id="sessionValue">--</span>
        </div>
    </div>
</div>

<script src="script.js"></script>
</body>
</html>

```

script.js

```

window.onload = function () {
    let savedTheme = localStorage.getItem("theme");

    if (savedTheme) {
        applyTheme(savedTheme);
        updateStorageDisplay();
        document.getElementById("message").innerText = "Welcome back! Theme restored.";
    }
};

function setTheme(theme) {
    applyTheme(theme);
    localStorage.setItem("theme", theme);
    sessionStorage.setItem("currentTheme", theme);
    updateStorageDisplay();
    document.getElementById("message").innerText = "Theme saved: " + theme;
}

function applyTheme(theme) {
    if (theme === "dark") {
        document.body.classList.add("dark");
    } else {
        document.body.classList.remove("dark");
    }
}

function clearTheme() {
    localStorage.removeItem("theme");
    sessionStorage.removeItem("currentTheme");
    document.body.classList.remove("dark");
    updateStorageDisplay();
    document.getElementById("message").innerText = "Theme preference cleared.";
}

function updateStorageDisplay() {
    let localVal = localStorage.getItem("theme") || "-";
    let sessionVal = sessionStorage.getItem("currentTheme") || "-";
    document.getElementById("localValue").innerText = localVal;
    document.getElementById("sessionValue").innerText = sessionVal;
}

```

4. Output:

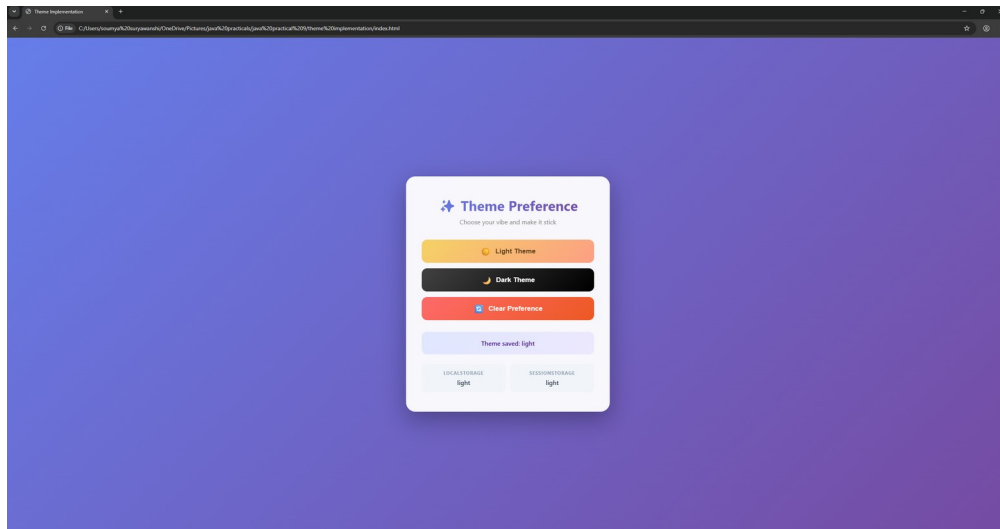


Fig 4: Theme Preference output — PRN: 24070521062, Name: Soumya Suryawanshi, File Path: C:/Users/soumya suryawanshi/OneDrive/Pictures/java practicals/java practical 209/theme implementation/index.html

5. Case Study Title:

Seminar Schedule Planner — Persisting Schedule Data and Session Selections with localStorage and sessionStorage

6. Case Study Program Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Seminar Schedule Planner</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <h1>Seminar Schedule Planner</h1>
    <p class="subtitle">Click any day to view topics & track your session</p>

    <div class="table-wrapper">
      <table>
        <thead>
          <tr>
            <th rowspan="3">Day</th>
            <th colspan="3">Seminar</th>
          </tr>
          <tr>
            <th colspan="2">Schedule</th>
            <th rowspan="2">Topic</th>
          </tr>
          <tr>
            <th>Begin</th>
            <th>End</th>
          </tr>
        </thead>
      </table>
    </div>
  </div>
</body>
</html>
```

```

        <th>End</th>
    </tr>
</thead>
<tbody>
    <tr class="day-row" data-day="Monday" onclick="selectDay('Monday')">
        <td class="day-cell">Monday</td>
        <td class="time-cell begin">8:00 a.m.</td>
        <td class="time-cell end">5:00 p.m.</td>
        <td>Introduction to XML<br>Validity: DTD and Relax NG</td>
    </tr>
    <tr class="day-row" data-day="Tuesday" onclick="selectDay('Tuesday')">
        <td class="day-cell" rowspan="3">Tuesday</td>
        <td class="time-cell begin">8:00 a.m.</td>
        <td class="time-cell end">11:00 a.m.</td>
        <td rowspan="2">XPath</td>
    </tr>
    <tr class="day-row" data-day="Tuesday" onclick="selectDay('Tuesday')">
        <td class="time-cell begin">11:00 a.m.</td>
        <td class="time-cell end">2:00 p.m.</td>
    </tr>
    <tr class="day-row" data-day="Tuesday" onclick="selectDay('Tuesday')">
        <td class="time-cell begin">2:00 p.m.</td>
        <td class="time-cell end">5:00 p.m.</td>
        <td>XSL Transformations</td>
    </tr>
    <tr class="day-row" data-day="Wednesday"
onclick="selectDay('Wednesday')">
        <td class="day-cell">Wednesday</td>
        <td class="time-cell begin">8:00 a.m.</td>
        <td class="time-cell end">12:00 p.m.</td>
        <td>XSL Formatting Objects</td>
    </tr>
</tbody>
</table>
</div>

<div class="status-card">
    <p id="message">No day selected yet</p>
</div>

<div class="storage-info">
    <div class="storage-item">
        <span class="storage-label">Last Visited (localStorage)</span>
        <span class="storage-value" id="localValue">—</span>
    </div>
    <div class="storage-item">
        <span class="storage-label">Session Clicks (sessionStorage)</span>
        <span class="storage-value" id="sessionValue">0</span>
    </div>
</div>

    <button class="btn-reset" onclick="clearData()">🔄 Clear All Data</button>
</div>

<script src="script.js"></script>
</body>
</html>

```

script.js

```
window.onload = function () {
  let lastDay = localStorage.getItem("lastViewedDay");
  let sessionClicks = sessionStorage.getItem("sessionClicks") || 0;

  if (lastDay) {
    highlightDay(lastDay);
    document.getElementById("message").innerText = "Welcome back! Last viewed: " + lastDay;
  }

  document.getElementById("localValue").innerText = lastDay || "-";
  document.getElementById("sessionValue").innerText = sessionClicks;
};

function selectDay(day) {
  let clicks = parseInt(sessionStorage.getItem("sessionClicks") || 0) + 1;
  sessionStorage.setItem("sessionClicks", clicks);
  localStorage.setItem("lastViewedDay", day);

  highlightDay(day);
  updateStorageDisplay();

  alert(" " + day + " Schedule Selected!\n\nSession Clicks: " + clicks);
}

function highlightDay(day) {
  document.querySelectorAll(".day-row").forEach(row => {
    row.classList.remove("active");
  });

  document.querySelectorAll('.day-row[data-day="' + day + '"]').forEach(row => {
    row.classList.add("active");
  });
}

function updateStorageDisplay() {
  let lastDay = localStorage.getItem("lastViewedDay") || "-";
  let sessionClicks = sessionStorage.getItem("sessionClicks") || 0;

  document.getElementById("localValue").innerText = lastDay;
  document.getElementById("sessionValue").innerText = sessionClicks;
}

function clearData() {
  localStorage.removeItem("lastViewedDay");
  sessionStorage.removeItem("sessionClicks");

  document.querySelectorAll(".day-row").forEach(row => {
    row.classList.remove("active");
  });

  updateStorageDisplay();
  document.getElementById("message").innerText = "All schedule data cleared.";
}
```

7. Output:

Day	Seminar		Topic
	Begin	End	
Monday	8:00 a.m.	5:00 p.m.	Introduction to XML, Velocity, CD and Role NG
	8:00 a.m.	11:00 a.m.	XPath
Tuesday	11:00 a.m.	2:00 p.m.	XSL Transformations
	2:00 p.m.	5:00 p.m.	XSL Formatting Objects
Wednesday	8:00 a.m.	12:00 p.m.	

No day selected yet

LAST VIEWED (LOCALSTORAGE): Monday SESSION CLICKS (SESSIONSTORAGE): 1

Clear All Data

Fig 7: Seminar Schedule Planner output — PRN: 24070521062, Name: Soumya Suryawanshi, File Path: C:/Users/soumya suryawanshi/OneDrive/Pictures/java practicals/java practical 209/case study/index.html

8. Result/Conclusion:

The experiment successfully demonstrates persistent and temporary browser-side storage using `localStorage` and `sessionStorage`. A Theme Preference page was implemented where the selected light or dark theme is saved to `localStorage` and `sessionStorage` and restored automatically on the next visit. In the accompanying case study, a Seminar Schedule Planner was built that persists schedule rows in `localStorage` and records the currently selected day and session-click count in `sessionStorage`, so the last-viewed day and click count remain available within the same browser session. Both implementations produced the expected output for the given inputs and reinforced practical understanding of the Web Storage API and its use in retaining user preferences and session state.

Upload the Soft copy as well as complete code on GitHub link is mandatory.